

Evidence Pack: AI for Grid Operations

Key Points

Efficient operation of the grid is dependent on the ability to predict what is about to happen and find the optimal set of actions to take. AI has significant potential to improve the efficiency and resilience of a future, more complex grid.

Prediction and optimisation are both becoming harder. This is because of more weather dependent generation technologies (wind/solar), decentralisation of energy technologies and the need for a more flexible demand side. Delivering a resilient grid in the face of increased complexity and uncertainty is already leading to rising balancing costs.

AI offers powerful tools to address this. While Large Language Models may support sector efficiency more widely, for grid operations it is AI advances in forecasting and optimisation that are most critical.

These are essential to underpin key capabilities of the future energy system, particularly flexibility and cost-effective integration of renewables. Failure to effectively embed AI usage in these areas will lead to higher than necessary balancing and reinforcement costs.

The overall scale of the potential benefits is difficult to quantify given the role AI can play involves mitigating increased complexity at the heart of system operation. The potential is huge, but uncertain. AI-based forecasting is already directly starting to release benefits, with more than £50m / year¹ in balancing cost savings.

For forecasting, the technology is relatively mature, and the challenge is adoption. DESNZ needs to make accelerating the adoption of AI-enabled forecasting in the sector a key goal. Openness is an important lever. Ofgem should encourage open publication and benchmarking of forecasts and work with NESO to establish forecasting markets that enable AI innovators to compete efficiently.

For AI-based optimisation, further R&D is necessary to successfully apply it to grid operations. Access to realistic grid operations data for AI training is a key barrier that could potentially be overcome using high-fidelity synthetic data and secure sandboxes.

Strategic context

As the grid becomes more decentralised, weather dependent and reliant on flexibility, it is becoming harder and more expensive to operate². Operational decisions involve 4 stages³:

- See: gather information to develop a clear understanding of the situation
- Think: evaluate different potential options and their consequences
- Act: implement the chosen actions
- Improve: learn from the effectiveness of past decisions.

While AI can be useful in all of those, as the complexity of the system increases it has the potential to be particularly transformative for See and Think⁴. Act on short timescales has value for system resilience⁵ but is less immediately critical.

Priority area: short-term forecasting and situational awareness

Operational decisions are made based on an assessment of the state of the grid now and over the next few minutes and hours. The potential for AI in the 'See' stage is integrating new data sources with better AI-driven short-term forecasting techniques to give a much more accurate assessment of what will happen next.

Why now?

AI-based forecasting has been advancing rapidly. New techniques mean AI can provide weather forecasts much faster⁶. This allows weather forecasts at much finer spatial and temporal granularities – which are key for energy – as well as many more possible scenarios to be considered.

Other AI time series forecasting techniques are becoming competitive⁷ (and easy to deploy). Increased data availability – for example synthetic or aggregated smart meter data – means forecasts that weren't possible a few years ago are possible now.

At the same time, the increase in renewable generation and flexibility have made forecasting energy supply and demand both harder and more important. Things we could reasonably do before with only basic forecasts will soon be impossible (or very expensive) without more accurate forecasts.

What are the benefits?

If we are more confident in what will happen in the next few hours we can take fewer actions earlier, which means we don't have to pay generators to be on standby or pay a premium for quicker responses.

There are significant opportunities to reduce both balancing and constraint costs that have either already been delivered or are in progress:

- AI-driven solar PV forecasting is driving estimated savings of £30m/year in reserve costs⁸
- AI-based weather forecasting and asset monitoring is saving £20m/year in constraint costs⁹ by increasing the power that can flow through overhead lines.
- £43m/year reduced balancing costs¹⁰ are expected to come from NESO's Volta Real Time Prediction project

Success Story: Solar Forecasting

Open Climate Fix and NESO combined cutting edge AI with satellite imagery, traditional weather forecasts and real-time PV readings to halve NESO's solar forecast errors. This led to a reduction in reserves held by the grid, driving estimated savings of £30m and 300k tCO₂e per year.

Better forecasts can also improve grid capacity in the short term through less conservative management of network assets¹¹.

The resilience and cost benefits will only grow. Forecasting is like the grease on the gears of the future energy system. If it's working well the whole system runs smoothly, but without it things quickly start to run inefficiently. We are adding many more gears to our system, so operating it in the future will be much harder without accurate forecasting.

Why isn't it happening faster?

The limiting factor isn't the capability of AI. Historically, access to network data has been a key obstacle. Significant progress has been made on that over the last few years, although timely publication of load data and incompleteness of network topology data¹² remain challenges.

The opportunity AI forecasting offers to better manage constraints at a local level is hard for distribution system operators to fully value due to limited local markets. This contrasts with national markets that make the financial benefits of improved forecasting relatively clear for NESO.

Success Story: Constraint Forecasting

UKPN proactively invested in building their AI forecasting capability and culture, including data pipelines, a forecasting platform, and AI forecasting experts.

Applying this to planned outages led to consumer benefits of £22m from 165GWh of generation not being constrained.

The need for changes to culture, processes and systems remain significant barriers. Some organisations like UKPN have been proactive in addressing that, but many have not. A key change that needs to be made is the shift to probabilistic forecasting – quantitatively combining the likelihood of different outcomes rather than using a handful of fixed scenarios.

What do we need to do?

DESNZ, DSIT, Ofgem and NESO need to treat short-term forecasting as a critical strategic capability of the energy sector that requires coordinated interventions to accelerate:

Innovate: technology companies and **DSIT** should bring the best and brightest minds together to further develop fast, high spatial and temporal resolution weather forecasting, which is very valuable for the energy system.

Open up: Ofgem should require organisations to publish forecasts and forecast accuracy in standardised formats that allow comparison and benchmarking.

Incentivise: NESO should trial an open forecast market (proposed by academics at Imperial¹³ and recently trialled in Belgium¹⁴) to enable competition in forecasting, automatic combination of the best forecasts, and rapid replacement of poorly performing forecasts.

Understand: DESNZ should develop a framework to understand in detail how improvements in different forecasts would translate into benefits for the system in 2030¹⁵.

Priority area: optimisation and decision support

The main way AI can add value to the 'Think' stage is by filtering through the vast number of potential combinations of futures and actions to present the most relevant to operators.

Why now?

The energy system is increasingly both distributed and bi-directional. As a result, the number of combinations that need to be considered to balance the system is increasing.

Existing methods of decision-making are creaking at the seams – for example, NESO has needed several major actions¹⁶ to increase the rate at which batteries are used by the grid and still has further to go.

There have been major strides in using AI to accelerate simulations in other fields. This could allow many more options to be considered than using traditional power systems simulations.

Reinforcement Learning has been rapidly improving in other fields, and it has the potential to be able to optimise more complex problems than the traditional optimisation techniques used in the energy sector.

What are the benefits?

As the energy system increases in complexity, our traditional approaches to balancing the system are becoming less effective¹⁷. The number of combinations that need to be considered to make optimal decisions exceeds the capacity of our more manual processes. This contributes to rising balancing costs¹⁸ and may gradually undermine our ability to maintain system resilience.

Whilst currently difficult to quantify, the key benefit of AI in this context is that it can evaluate vastly more possibilities than manual processes and present the most important ones to operators for a decision.

For example, the NESO Volta Grand Optimiser Design¹⁹ project (in collaboration with Alphabet's moonshot innovation lab) is looking at how AI better optimise both day-ahead and within-day decisions given varying uncertainty. This should mean operators will be able to take fewer, cheaper balancing actions. The scale of the potential benefits to consumers has yet to be quantified by the programme.

The Learn to Run a Power Network competition has demonstrated a 75% reduction in simulated redispatch costs²⁰ in artificial networks when using AlphaZero.

In the Netherlands, Tennet have run a pilot of GridOptions²¹ to optimise network configuration in a small part of their network. They estimate a 10-20% reduction in congestion²² is achievable.

Why isn't it happening faster?

NESO have been doing good work in this space in the last two years – the NESO Volta programme is working to lay out a clear roadmap that will deliver change effectively.

However, France and the Netherlands invested 5-7 years ago in developing research centres and academia and industry partnerships focused on reinforcement learning for grid control. They have made good progress in developing tools and frameworks and ran public competitions (e.g. Learn to Run a Power Network²³). As a result, they have attracted talent and built a community around their tools. The UK has the opportunity to develop a complimentary initiative leveraging British data and expertise that would provide innovators and researchers with access to the critical data needed to accelerate R&D.

The national security considerations around system operation data have severely limited the ability for AI researchers to develop and refine techniques that are tailored for the challenges faced in grid operation. This is exacerbated by the fact that the manual nature of existing control room processes means lots of useful data about decision making is not automatically captured in sufficient detail for AI systems to learn effectively from it.

Realising the benefits of AI systems will require substantial process and cultural change within the system operation community. The pace of AI adoption will likely be as limited by the pace of this change as by technological capability.

What do we need to do?

Realising the benefits from AI optimisation sooner rather than later requires a two-pronged approach: a strong drive from the centre to shape the required process and cultural change, and with wider access to data to enable innovation to thrive.

Programme: NESO should continue to prioritise the Volta programme as a critical initiative to modernise control room operations, resourcing it with top internal and external talent and focusing on process and culture change as well as technology.

Synthetic data: UKRI should fund a NESO-supported programme to create openly available simulation environments and synthetic data that capture the key dynamics of GB system operation sufficiently to allow effective AI R&D for the GB system.

Data: Ofgem and NESO should work together to provide secure access to system operations data for AI training to AI researchers and developers through the REVEAL²⁴ platform or an Ofgem AI sandbox.

Research: UKRI, DESNZ and Ofgem should co-ordinate to ensure funding mechanisms focused on research on AI for grid optimisation can include guaranteed access to systems operation data (synthetic or real).

Appendix: additional actions to accelerate AI adoption

DSIT should:

- Initiate a project to collate existing granular weather observation datasets from diverse proprietary sources (e.g. wind speed measurements done by wind turbine operators) across the UK for use in training and calibrating high resolution AI weather forecasts.

DESNZ and DSIT should:

- Consider partnering with a technology company to establish an academic research lab specifically focused on reinforcement learning for grid operations.

Ofgem should:

- Host (or fund) an open platform that enables direct comparison of distribution network forecast accuracies²⁵ to incentivise improvement.
- Require or incentivise sector participants (especially DNOs) to invest in forecasting capability, produce specific forecasts and monitor accuracy.
- Accelerate realisation of the benefits from existing AI forecasting tools by making someone responsible for getting Dynamic Line Ratings rolled out faster and delivering benefits (internal industry estimates suggest there is potentially £50m+ per year in constraint cost savings to be had from rollout on known constrained circuits).
- Consider strengthening incentives around network data provision that supports AI even further, for example by making access to AI-related innovation funding for networks conditional on their successful completion of data-related milestones in their Digitalisation Strategy and Action Plans.

NESO should:

- Map the sector-wide forecasting capabilities that will be required to operate the system efficiently in 2030 and develop a roadmap to achieve those.
- Publish existing short-term forecast detailed methodologies and code openly.
- Ensure data access initiatives like REVEAL²⁶ are designed to enable AI-based forecasting innovation.
- Embed deep technical skills in forecasting in NESO (and DSOs) through academic and deep tech collaborations.
- Ensure redesigned control room processes and systems collect the right data for future AI training.
- Define (and publish) a list of the datasets that are critical to system operations and launch an initiative to clean them up and bring them together for AI training.

References

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